

Computer Architecture Lab (Week 1)

Introduction



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- ❑ Research Interest: AI, NLP
 - ❑ 2018-2023: Engineer's Degree in Computer Science at Institute of Technology of Cambodia and currently master student at CADT.

Course Evaluation

☐ Course	15 h
☐ Practice	30 h
☐ Attendance:	10%
☐ Homework/Quiz:	10%
☐ Project:	20%
☐ Mid-term Exam:	20%
☐ Final Exam:	40%

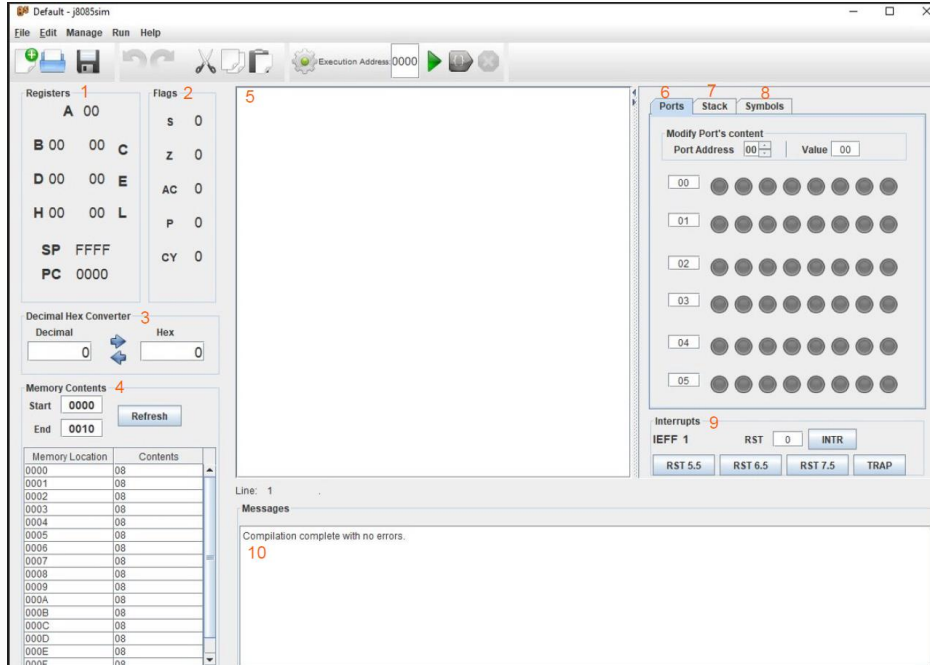
Schedule

Week 1	Week 2	Week 3	Week 4	Week 5
I. Assembly Language Practice: • Introduction to assembly language • Practice with exercises	I. Assembly Language Practice: • Practice with exercises	I. Assembly Language Practice: • Practice with exercises	I. Assembly Language Practice: • Practice with exercises	I. Assembly Language Practice: • Practice with exercises

Schedule

Week 6	Week 7	Week 8	Week 9	Week 10
I. Assembly Language Practice (1h: 30mn) II. Mid-Term (1h: 30mn)	I. Logic: • Introduction • Practice with exercises • Group Division	I. Logic • Practice with exercises • Project Progress update	I. Logic • Practice with exercises • Project Progress update	I. Logic • Final Presentation

Introduction to Assembly language IDE



I. Download Tool

- Requirement: Download Java Runtime Environment (JRE) with this link <https://www.java.com/en/download/manual.jsp> and open IDE with JRE.

II. Introduction to the tool

- Register: A= Accumulator, B,C, D, E, H, L register.
- Flage: S = sign Z = zero, AC = Auxiliary Carry, P = Parity, CY = Carry
- Converter: Used to convert between decimal and hexadecimal values.
- Memory Content: Shows memory address vs contents.
- Code Editor Window: Where you write 8085 assembly language.
- Port: Used to view and modify I/O port contents.
- Stack; Displays the stack contents when using PUSH/POP instructions.
- Symbols: Displays symbols and labels used in the assembly program.
- Interrupts: TRAP, RST 7.5, RST 6.5, RST 5.5, INTR
- Message: Shows compiler messages, errors, and program execution results.

Classification of Instruction Set

- ☐ Data Transfer Instruction: Move, copy, load, store, exchange data
- ☐ Arithmetic Instructions: Add, Subtract, Increase, decrease complement
- ☐ Logical Instructions: AND, OR, and XOR
- ☐ Branching Instructions: Jump, call, return
- ☐ Control Instruction: Push, pop, disable, enable

Data Transfer Instructions

- These instructions move data between registers, or between memory and registers.
- These instructions copy data from source to destination.
- While copying, the contents of source are not modified.

Data Transfer Instructions

Opcode	Operand	Description
MOV	Rd, Rs M, Rs Rd, M	Copy from source to destination.

- This instruction copies the contents of the source register into the destination register.
- The contents of the source register are not altered.
- If one of the operands is a memory location, its location is specified by the contents of the HL registers.
- **Example:** MOV B, C or MOV B, M

Data Transfer Instructions

Opcode	Operand	Description
MVI	Rd, Data M, Data	Move immediate 8-bit

- The 8-bit data is stored in the destination register or memory.
- If the operand is a memory location, its location is specified by the contents of the H-L registers.
- **Example:** MVI B, 57H or MVI M, 57H

Data Transfer Instructions

Opcode	Operand	Description
LDA	16-bit address	Load Accumulator

- The contents of a memory location, specified by a 16-bit address in the operand, are copied to the accumulator.
- The contents of the source are not altered.
- **Example:** LDA 2034H

Data Transfer Instructions

Opcode	Operand	Description
LDAX	B/D Register Pair	Load accumulator indirect

- The contents of the designated register pair point to a memory location.
- This instruction copies the contents of that memory location into the accumulator.
- The contents of either the register pair or the memory location are not altered.
- **Example:** LDAX B

Data Transfer Instructions

Opcode	Operand	Description
LXI	Reg. pair, 16-bit data	Load register pair immediate

- This instruction loads 16-bit data in the register pair.
- **Example:** LXI H, 2034 H

Data Transfer Instructions

Opcode	Operand	Description
LHLD	16-bit address	Load H-L registers direct

- This instruction copies the contents of memory location pointed out by 16-bit address into register L.
- It copies the contents of next memory location into register H.
- **Example:** LHLD 2040 H

Data Transfer Instructions

Opcode	Operand	Description
STA	16-bit address	Store accumulator direct

- The contents of accumulator are copied into the memory location specified by the operand.
- **Example:** STA 2500 H

Data Transfer Instructions

Opcode	Operand	Description
STAX	Reg. pair	Store accumulator indirect

- The contents of accumulator are copied into the memory location specified by the contents of the register pair.
- **Example:** STAX B

Data Transfer Instructions

Opcode	Operand	Description
SHLD	16-bit address	Store H-L registers direct

- The contents of register L are stored into memory location specified by the 16-bit address.
- The contents of register H are stored into the next memory location.
- **Example:** SHLD 2550 H

Data Transfer Instructions

Opcode	Operand	Description
XCHG	None	Exchange H-L with D-E

- The contents of register H are exchanged with the contents of register D.
- The contents of register L are exchanged with the contents of register E.
- **Example:** XCHG

Data Transfer Instructions

Opcode	Operand	Description
SPHL	None	Copy H-L pair to the Stack Pointer (SP)

- This instruction loads the contents of H-L pair into SP.
- **Example:** SPHL

Data Transfer Instructions

Opcode	Operand	Description
XTHL	None	Exchange H-L with top of stack

- The contents of L register are exchanged with the location pointed out by the contents of the SP.
- The contents of H register are exchanged with the next location (SP + 1).
- **Example:** XTHL

Data Transfer Instructions

Opcode	Operand	Description
PCHL	None	Load program counter with H-L contents

- The contents of registers H and L are copied into the program counter (PC).
- The contents of H are placed as the high-order byte and the contents of L as the low-order byte.
- **Example:** PCHL

Data Transfer Instructions

Opcode	Operand	Description
PUSH	Reg. pair	Push register pair onto stack

- The contents of register pair are copied onto stack.
- SP is decremented and the contents of high-order registers (B, D, H, A) are copied into stack.
- SP is again decremented and the contents of low-order registers (C, E, L, Flags) are copied into stack.
- **Example:** PUSH B

Data Transfer Instructions

Opcode	Operand	Description
POP	Reg. pair	Pop stack to register pair

- The contents of top of stack are copied into register pair.
- The contents of location pointed out by SP are copied to the low-order register (C, E, L, Flags).
- SP is incremented and the contents of location are copied to the high-order register (B, D, H, A).
- **Example:** POP H

Data Transfer Instructions

Opcode	Operand	Description
OUT	8-bit port address	Copy data from accumulator to a port with 8-bit address

- The contents of accumulator are copied into the I/O port.
- **Example:** OUT 78 H

Data Transfer Instructions

Opcode	Operand	Description
IN	8-bit port address	Copy data to accumulator from a port with 8-bit address

- The contents of I/O port are copied into accumulator.
- **Example:** IN 8C H

Addition

- Any 8-bit number, or the contents of register, or the contents of memory location can be added to the contents of accumulator.
- The result (sum) is stored in the accumulator.
- No two other 8-bit registers can be added directly.
- **Example:** The contents of register B cannot be added directly to the contents of register C.

Subtraction

- Any 8-bit number, or the contents of register, or the contents of memory location can be subtracted from the contents of accumulator.
- The result is stored in the accumulator.
- Subtraction is performed in 2's complement form.
- If the result is negative, it is stored in 2's complement form.
- No two other 8-bit registers can be subtracted directly.

Increment / Decrement

- The 8-bit contents of a register or a memory location can be incremented or decremented by 1.
- The 16-bit contents of a register pair can be incremented or decremented by 1.
- Increment or decrement can be performed on any register or a memory location.

Arithmetic Instructions

Opcode	Operand	Description
ADD	R M	Add register or memory to accumulator

- The contents of register or memory are added to the contents of accumulator.
- The result is stored in accumulator.
- If the operand is memory location, its address is specified by H-L pair.
- All flags are modified to reflect the result of the addition.
- **Example:** ADD B or ADD M

Arithmetic Instructions

Opcode	Operand	Description
ADC	R M	Add register or memory to accumulator with carry

- The contents of register or memory and Carry Flag (CY) are added to the contents of accumulator.
- The result is stored in accumulator.
- If the operand is memory location, its address is specified by H-L pair.
- All flags are modified to reflect the result of the addition.
- **Example:** ADC B or ADC M

Arithmetic Instructions

Opcode	Operand	Description
ADI	8-bit data	Add immediate to accumulator

- The 8-bit data is added to the contents of accumulator.
- The result is stored in accumulator.
- All flags are modified to reflect the result of the addition.
- **Example:** ADI 45 H

Arithmetic Instructions

Opcode	Operand	Description
ACI	8-bit data	Add immediate to accumulator with carry

- The 8-bit data and the Carry Flag (CY) are added to the contents of accumulator.
- The result is stored in accumulator.
- All flags are modified to reflect the result of the addition.
- **Example:** ACI 45 H

Arithmetic Instructions

Opcode	Operand	Description
DAD	Reg. pair	Add register pair to H-L pair

- The 16-bit contents of the register pair are added to the contents of H-L pair.
- The result is stored in H-L pair.
- If the result is larger than 16 bits, then CY is set.
- No other flags are changed.
- **Example:** DAD B

Arithmetic Instructions

Opcode	Operand	Description
SUB	R M	Subtract register or memory from accumulator

- The contents of the register or memory location are subtracted from the contents of the accumulator.
- The result is stored in accumulator.
- If the operand is memory location, its address is specified by H-L pair.
- All flags are modified to reflect the result of subtraction.
- **Example:** SUB B or SUB M

Arithmetic Instructions

Opcode	Operand	Description
SBB	R M	Subtract register or memory from accumulator with borrow

- The contents of the register or memory location and Borrow Flag (i.e. CY) are subtracted from the contents of the accumulator.
- The result is stored in accumulator.
- If the operand is memory location, its address is specified by H-L pair.
- All flags are modified to reflect the result of subtraction.
- **Example:** SBB B or SBB M

Arithmetic Instructions

Opcode	Operand	Description
SUI	8-bit data	Subtract immediate from accumulator

- The 8-bit data is subtracted from the contents of the accumulator.
- The result is stored in accumulator.
- All flags are modified to reflect the result of subtraction.
- **Example:** SUI 45 H

Arithmetic Instructions

Opcode	Operand	Description
SBI	8-bit data	Subtract immediate from accumulator with borrow

- The 8-bit data and the Borrow Flag (i.e. CY) is subtracted from the contents of the accumulator.
- The result is stored in accumulator.
- All flags are modified to reflect the result of subtraction.
- **Example:** SBI 45 H

Arithmetic Instructions

Opcode	Operand	Description
INR	R M	Increment register or memory by 1

- The contents of register or memory location are incremented by 1.
- The result is stored in the same place.
- If the operand is a memory location, its address is specified by the contents of H-L pair.
- **Example:** INR B or INR M

Arithmetic Instructions

Opcode	Operand	Description
INX	R	Increment register pair by 1

- The contents of register pair are incremented by 1.
- The result is stored in the same place.
- **Example:** INX H

Arithmetic Instructions

Opcode	Operand	Description
DCR	R M	Decrement register or memory by 1

- The contents of register or memory location are decremented by 1.
- The result is stored in the same place.
- If the operand is a memory location, its address is specified by the contents of H-L pair.
- **Example:** DCR B or DCR M

Arithmetic Instructions

Opcode	Operand	Description
DCX	R	Decrement register pair by 1

- The contents of register pair are decremented by 1.
- The result is stored in the same place.
- **Example:** DCX H

1. Store 8-bit data in memory

- ❑ Store the data byte 32H into memory location 4000H

2. Exchange the contents of memory locations

- ❑ Statement: Exchange the contents of memory locations 4000H and 4001H.

3. Add two 8-bit numbers

- ❑ Statement: Add the contents of memory locations 4000H and 4001H and place the result in memory location 4002H.

4. Subtract two 8-bit numbers

- ❑ Statement: Subtract the contents of memory locations 4200H and 4201H and place the result in memory location 4202H.

5. Add contents of two memory locations

- ☐ Statement: Add the contents of memory locations 4001H and 4002H
- ☐ Place the result in the memory locations 4003H and 4004H.
- ☐ The most significant byte in location 4004H

6. Finding one's complement of a number

- ☐ Statement:
- ☐ Find the 1's complement of the number stored at memory location 4400H
- ☐ Store the complemented number at memory location 4300H.

7. Finding Two's complement of a number

- ☐ Statement: Find the 2's complement of the number stored at memory location 4200H
- ☐ Store the complemented number at memory location 4300H.

8. Add two 16-bit numbers

- ☐ Statement:
- ☐ Add the 16-bit number in memory locations 4000H and 4001H to the 16-bit number in memory locations 4002H and 4003H.
- ☐ The most significant eight bits of the two numbers to be added are in memory locations 4001H and 4003H.
- ☐ Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.

Thanks